Day: Solved Past Paper

Date:

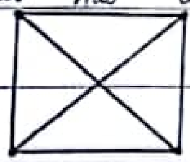
(Discrete Structure)

Bs 2nd term (CS) 2020

Short Questions

* **Define Complete Graph?**

A complete graph is a graph in which every vertex is connected to every other vertex. In other words each pair of vertices has an edge between them. ∴ K_4 is a complete graph.



* **Define Bipartite Graph?**

A bipartite graph is a graph whose vertices can be divided into two disjoint sets, U and V, such that every edge connects a vertex in U to a vertex V. In other words, there are no edges between vertices within the same set. ∴ $K_{2,3}$ is a complete Bipartite graph.

* **Evaluate these quantities -17 mod 2?**

$-17 \bmod 2 = 1$

Because -17 divided by 2 leaves a remainder of 1

* **Explain Prims algorithm?**

Prims Algorithm is a greedy ~~a~~ algorithm use to find the minimum spanning tree of a connected, undirected and weighted graph. It works by ~~a~~ selecting the smallest edge that connects

Day: Date:

a vertex in the tree to a vertex not yet in the tree

* **Explain Tautology?**

A tautology is a compound proposition that is always true, regardless of the truth values of its component propositions. For Example, "$P \vee \neg P$" is a tautology because it is always ~~true~~ true

* **State Inclusion - Exclusion principle?**

The inclusion-exclusion principle is a counting principle that states that for two ~~f~~ sets A and B, the size of the union of A and B is equal to the sum of the sizes of A and B, minus the size of their intersection.

* **When do you say that two compound propositions are logically equivalent?**

Two compound propositions are logically equivalent if they have the same truth value for every possible combination of truth values of their compound propositions

* **What is the term a_8 of the sequence $\{a_n\}$ if a_n equals 2^{n-1}?**

Given that $a(n) = 2^{n-1}$

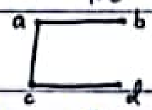
We want to find $a(s)$.
Since $a(n) = 2^{n-1}$, we can substitute s for n.

$a(s) = 2^{s-1}$

So, the term $a(s)$ of the sequence $\{a(n)\}$ is 2^{s-1}.

* **Define Spanning tree?**

A spanning tree of a connected, undirected graph is a subgraph that is a tree and includes all the vertices of the original graph. eg:



* **What are the Basic Counting Principles?**

The Basic Counting Principles include the multiplication principle, the addition principle, and the inclusion-exclusion principle.

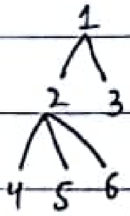
* **What is the power set of the set $\{1, 3, 5\}$?**

The power set of $\{1, 3, 5\}$ is $\{\{\}, \{1\}, \{3\}, \{5\}, \{1,3\}, \{1,5\}, \{3,5\}, \{1,3,5\}\}$

* **Define the Height of a tree?**

The height of the tree is the maximum number of edges between the root node and any leaf node. In other words it is the longest path from the root node to a leaf node

For Example,



* **What is the value of $\Sigma_{j=1}^{5} j^2$?**

If we want to start summation from 0 we replace j by $j+1$.

$\Sigma_{j+1=1}^{5} (j+1)^2$

$\Sigma_{j=0}^{5} (j+1)^2$

$= 1^2 + 2^2 + 3^2 + 4^2 + 5^2$

$= 1 + 4 + 9 + 16 + 25$

$= 55$

* **Define Big-O notation?**

Big-O notation is a mathematical notation that describes the upper bound of an algorithm's time or space complexity, usually expressed as a function of the size of the input. Example, Algorithm: Find an element in an array.

Time Complexity: $f(n) = 2n + 1$

Big-O notation: $O(n)$

Meaning: The alogrithm's running time grows linearly with the size of the input array.

Discrete Structure (BS 2nd Term 2024)

Q Write short answers of following

ii) Define a wheelingraph.

Ans Wheel graph is denoted by W_n (wheel of size n) $n>=3$: the graph obtained from C_n by adding a vertex and edges from this vertex to the original vertex in C_n.

v) Define Simple graph.

Ans A Simple graph is an undirected graph with no multiple edges or loops.

vii) What is a loop in graph theory?

Ans In a graph theory we refers to loop as an edge ~~containing~~ connecting a vertex with itself.

viii) What is a bipartite graph?

Ans A graph with vertex set that can be partitioned into subsets V_1 and V_2 such that each edge connects a vertex in V_1 and a vertex in V_2.

ix) What does the notation $K_{m,n}$ represents in a graph? Meaning of $K_{m,n}$ is Complete bipartite graph

Ans The graph with vertex set partitioned into a subset of m elements and a subset of n elements such that two vertices are connected by an edge if and only if one is in the first subset and the other is in second subset.

x. What is a bijective function?

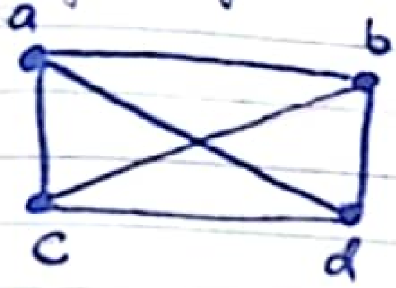
Ans A function is a bijective function if it is both onto and one-to-one function. for example: $\{a,b,c,d\}$ to $\{1,2,3,4\}$ with $f(a)=4$, $f(b)=2$, $f(c)=1$ and $f(d)=3$. It is a bijective function.

xi. Is the inverse of one-to-one function is also a function? Give one example in favour of your answer?

Ans Yes, the inverse of one to one function is also a function because we know The inverse of $f^{-1}(x)$ is $f(x)$ according to this one-to-one function also have inverse $f^{-1}(x)$ is one-to-one. $f(x)=x+1$

for example:- The function f has an inverse because it is a one to one we have to show it's inverse is one to one if we take inverse $f^{-1}(y)=y-1$ which is also a one to one because $y-1$ has the unique element of Z that is sent to y by f.

ii. If there is a complete graph with 4-vertices how many edges it will contain?



Solution A complete graph with 4 vertices have 6 edges according to hand shaking theorm put value of deg

$$2e = deg(v)$$

$$e = \frac{3 \times 4}{2} = e = \frac{12}{2} \quad \boxed{e = 6}$$

ii. Define De-Morgan's law in set theory?

Ans In set theory, De Morgan's law state that the complement of the union of two sets is equal to the intersection of their complements and the complement of the intersection of two sets is equal to the union of their intersection.

$$(A \cup B)' = A' \cap B'$$

$$(A \cap B)' = A' \cup B'$$

iii What is conjunction in rule of inference

iv. Define tautology?

v. What is truth table of $\forall x \exists y\, P(x,y)$ where x and y are set of real number.